

Application of the Riccati Equation & Design of an LQR for a Single-Axis Missile Pitch Model

1 System Dynamics and Cost Functional Formulations

Consider a continuous-time linear time-invariant (LTI) dynamic system governed by the state-space equation:

$$\dot{x}(t) = Ax(t) + Bu(t) \quad (1)$$

To design an optimal control policy over a finite time horizon $t \in [t_0, t_f]$, a quadratic performance index J is defined in scalar-matrix form:

$$J = \frac{1}{2}x(t_f)^T Q_f x(t_f) + \frac{1}{2} \int_{t_0}^{t_f} (x(t)^T Q(t)x(t) + 2x(t)^T N(t)u(t) + u(t)^T R(t)u(t)) dt \quad (2)$$

For infinite-horizon regulator problems where $t_f \rightarrow \infty$, terminal state constraints are relaxed, yielding the stationary performance functional:

$$J = \frac{1}{2} \int_0^\infty (x(t)^T Q(t)x(t) + 2x(t)^T N(t)u(t) + u(t)^T R(t)u(t)) dt \quad (3)$$

Physical Interpretations of Cost Matrix Terms

- $x(t_f)^T Q_f x(t_f)$ — **Terminal Cost Weighting**: Penalizes residual state deviations at the final time t_f , ensuring precise trajectory completion.
- $u(t)^T R(t)u(t)$ — **Control Expenditure Penalty**: Penalizes control effort to prevent actuator saturation and reduce physical energy consumption.
- $2x(t)^T N(t)u(t)$ — **Cross-Coupling Penalty**: Penalizes specific joint interactions between system states and control inputs.

Solvability Constraints

To guarantee a well-posed optimization problem with a unique, stable solution, the composite weighting matrix must be positive semi-definite, and the control penalty matrix must be strictly positive definite:

$$\begin{bmatrix} Q & N \\ N^T & R \end{bmatrix} \geq 0 \quad \text{and} \quad R > 0 \quad (4)$$

2 Algebraic Riccati Equation and Numerical Solvers

For infinite-horizon Linear Quadratic Regulator (LQR) problems, the optimal state-feedback gain matrix is derived from the symmetric positive semi-definite solution matrix P satisfying the Continuous Algebraic Riccati Equation (CARE):

$$A^T P + PA + Q - PBR^{-1}B^T P = 0 \quad (5)$$

Once the unique stabilizing solution matrix P is calculated, the optimal state-feedback control law $u(t) = -Kx(t)$ is established to stabilize the closed-loop system.

2.1 Numerical Solution Methods for CARE

2.1.1 Eigen-Decomposition Method (Direct Solvers)

This approach converts the non-linear quadratic matrix equation of dimension $n \times n$ into a linear $2n \times 2n$ matrix problem governed by the system Hamiltonian matrix H :

$$H = \begin{bmatrix} A & -BR^{-1}B^T \\ -Q & -A^T \end{bmatrix} \quad (6)$$

2.1.2 Laub's Schur Method

Direct eigenvector computation can suffer from severe numerical instability in the presence of repeated or ill-conditioned eigenvalues. Laub's Schur method provides a numerically robust alternative by employing an ordered real Schur decomposition of H :

1. Compute the real Schur decomposition $H = UTU^T$, where T is quasi-upper triangular and $U = \begin{bmatrix} U_{11} & U_{12} \\ U_{21} & U_{22} \end{bmatrix}$ is orthogonal.
2. Reorder T using orthogonal Givens rotations such that the n stable eigenvalues satisfying $\text{Re}(\lambda) < 0$ occupy the upper-left $n \times n$ block of T .
3. Extract the stable invariant subspace spanned by the first n columns of U to compute P :

$$P = U_{21}U_{11}^{-1} \quad (7)$$

3 Missile Pitch Dynamics and Environmental Modeling

3.1 Atmospheric and Physical Parameters

The physical geometry and atmospheric operating conditions governing missile pitch dynamics are modeled as follows:

- **Pitch Moment of Inertia:** $I_{yy} = \frac{1}{12}ML^2 + \frac{1}{4}MR^2$
- **Ambient Temperature Profile:**

$$T = \begin{cases} 216.65 \text{ K}, & \text{if altitude} > 11 \text{ km} \\ 288.15 - 0.0065 \times \text{altitude}, & \text{if altitude} \leq 11 \text{ km} \end{cases} \quad (8)$$

- **Speed of Sound:** $a = \sqrt{\gamma RT}$
- **Flight Velocity:** $V = \text{Mach number} \times a$
- **Atmospheric Pressure:** $p = 101325 \times e^{-h/H}$, $H = 8500 \text{ m}$
- **Air Density:** $\rho = 1.225 \times e^{-h/H}$
- **Dynamic Pressure:** $\bar{q} = \frac{1}{2}\rho V^2$

3.2 Dimensional Stability Derivatives

Normalizing dynamic parameters by missile mass m , reference area S , mean aerodynamic chord $\bar{c}$, and velocity V yields the dimensional stability derivatives:

$$Z_\alpha = \frac{\bar{q}S}{mV}C_{Z\alpha}, \quad Z_\delta = \frac{\bar{q}S}{mV}C_{Z\delta}, \quad M_\alpha = \frac{\bar{q}S\bar{c}}{I_{yy}}C_{m\alpha}, \quad M_\delta = \frac{\bar{q}S\bar{c}}{I_{yy}}C_{m\delta} \quad (9)$$

3.3 State-Space Representation

Choosing the state vector as $x(t) = [\Delta\alpha(t), \Delta q(t)]^T$ and control input as $u(t) = \Delta\delta(t)$, the linearized system dynamics are formulated as:

$$A = \begin{bmatrix} Z_\alpha & 1 + Z_q \\ M_\alpha & M_q \end{bmatrix}, \quad B = \begin{bmatrix} Z_\delta \\ M_\delta \end{bmatrix} \quad (10)$$

4 Linearization and Aerodynamic Derivatives

4.1 Bryson's Rule for Cost Function Scaling

To account for unit discrepancies across state and control variables, Bryson's rule scales each diagonal entry in Q and R by the inverse square of its maximum allowable physical limit (with angular units expressed in radians):

$$Q = \begin{bmatrix} \frac{1}{\alpha_{\max}^2} & 0 \\ 0 & \frac{1}{q_{\max}^2} \end{bmatrix}, \quad R = \begin{bmatrix} \frac{1}{\delta_{\max}^2} \end{bmatrix} \quad (11)$$

4.2 Linearization Around a Trim Condition

Missile flight dynamics are non-linear. To design a linear quadratic controller, the non-linear equations of motion are linearized about a trimmed flight equilibrium condition (x_0, u_0) using a first-order Taylor series expansion:

$$\dot{x} \approx f(x_0, u_0) + \left. \frac{\partial f}{\partial x} \right|_0 \Delta x + \left. \frac{\partial f}{\partial u} \right|_0 \Delta u \quad (12)$$

Short-Period Mode Analysis: The pitch channel response is dominated by the short-period mode, characterized by rapid rotational pitching and fast variation in angle of attack α . Matrix A captures how the system state rates $(\dot{\alpha}, \dot{q})$ evolve due to state perturbations independent of control input.

4.2.1 Pitch Acceleration ($\dot{q}$)

From Newton's second law for rotational dynamics about the missile pitch axis (Y_B):

$$I_{yy}\dot{q} = M_{\text{total}}(\alpha, q, \delta) \quad (13)$$

Linearizing about the trim point yields:

$$\Delta\dot{q} = \underbrace{\left(\frac{1}{I_{yy}} \frac{\partial M}{\partial \alpha} \right)}_{M_\alpha} \Delta\alpha + \underbrace{\left(\frac{1}{I_{yy}} \frac{\partial M}{\partial q} \right)}_{M_q} \Delta q \quad (14)$$

- $M_\alpha \rightarrow$ **Pitch Stiffness / Static Longitudinal Stability:** Quantifies the restoring aerodynamic torque per unit change in α .
 - $M_\alpha < 0 \implies$ Statically stable (generates a stabilizing restoring moment toward zero angle of attack).
 - $M_\alpha > 0 \implies$ Aerodynamically unstable (requires active feedback augmentation).
- $M_q \rightarrow$ **Pitch Damping Derivative:** Quantifies aerodynamic resistance opposing pitch rate rotation.

4.2.2 Angle-of-Attack Rate ($\dot{\alpha}$)

The rate of change of angle of attack is derived from fundamental kinematic relations along the body vertical axis (Z_B). Kinematically:

$$\theta = \gamma + \alpha \implies \dot{\alpha} = \dot{\theta} - \dot{\gamma} = q - \dot{\gamma} \quad (\text{where } \gamma = \text{flight path angle}) \quad (15)$$

Applying Newton's second law along the lift vector axis, the flight path angle rate $\dot{\gamma}$ is driven by vertical aerodynamic force F_z :

$$mV\dot{\gamma} = -F_z \implies \dot{\gamma} = -\frac{F_z}{mV} \quad (16)$$

5 Kinematic Relationships and Cost Functional Tuning

Combining the expressions for flight path rate and kinematics yields the non-linear angle-of-attack derivative:

$$\dot{\alpha} = q + \frac{F_z(\alpha, q, \delta)}{mV} \quad (17)$$

Linearizing the aerodynamic force contribution gives:

$$\frac{F_z}{mV} \approx \underbrace{\left(\frac{1}{mV} \frac{\partial F_z}{\partial \alpha} \right)}_{Z_\alpha} \Delta\alpha + \underbrace{\left(\frac{1}{mV} \frac{\partial F_z}{\partial q} \right)}_{Z_q} \Delta q \quad (18)$$

$$\implies \Delta\dot{\alpha} = Z_\alpha \Delta\alpha + (1 + Z_q) \Delta q \quad (19)$$

Physical Interpretation of Terms

- **Kinematic Coupling (1):** Pure body rotation. In the absence of spatial velocity vector reorientation ($\dot{\gamma} = 0$), pitching at rate q increases α at a 1 : 1 ratio.
- Z_α (**Force Derivative**): Vertical aerodynamic force generated per unit variation in α . Under standard sign conventions (Z -axis pointing downward), lift generation makes Z_α strictly negative.
- Z_q (**Rotary Force Derivative**): Incremental normal force generated across aerodynamic surfaces during pitching rotation ($Z_q \ll 1$).
- M_δ (**Control Effectiveness**): Control moment per unit fin deflection relative to inertia I_{yy} ; serves as the primary control authority metric for pitch rotational acceleration.
- Z_δ (**Direct Control Force Derivative**): Direct normal force generated immediately upon deflecting the control surface.

Note on Non-Minimum-Phase Behavior: Deflecting a tail control fin to initiate an upward pitch initially exerts a downward normal force (Z_δ). This creates a transient initial drop in trajectory prior to angle-of-attack buildup, producing non-minimum-phase dynamics (right-half-plane zeros).

5.1 Cost Functional Tuning Strategies

The explicit quadratic cost functional is expressed as:

$$J = \int_0^\infty (x^T Q x + u^T R u) dt = \int_0^\infty (Q_{11} \Delta\alpha^2 + Q_{22} \Delta q^2 + R_{11} \Delta\delta^2) dt \quad (20)$$

Selecting unweighted identity matrices ($Q = I$, $R = I$) introduces severe scale distortion, as numerical values for pitch rates Δq typically dwarf angle-of-attack perturbations $\Delta\alpha$. Applying Bryson's normalization ($Q_{11} = \frac{1}{\Delta\alpha_{\max}^2}$) scales non-dimensional errors into the unit range $[0, 1]$.

- **Increasing Q_{11} :** Heavily penalizes angle-of-attack deviations, prioritizing strict trajectory tracking at the expense of aggressive control surface activity and higher actuator slew rates.
- **Increasing R_{11} :** Heavily penalizes actuator movement, promoting smooth fin deflections and preventing actuator fatigue at the expense of slower transient tracking response.

6 Aerodynamic Coefficient Definitions

- **C_Z (Vertical Force Coefficient):** Dimensionless aerodynamic force component acting along the vehicle's body Z -axis.
- **C_m (Pitching Moment Coefficient):** Dimensionless aerodynamic pitching torque about the center of gravity.
- **$C_{Z\alpha}$ (Normal Force Curve Slope):** Sensitivity of vertical force coefficient with respect to angle of attack (α).
- **$C_{Z\delta}$ (Control Surface Force Coefficient):** Sensitivity of vertical force coefficient with respect to control surface deflection (δ).
- **$C_{m\alpha}$ (Static Longitudinal Stability Derivative):** Aerodynamic stiffness coefficient. A negative sign ($C_{m\alpha} < 0$) signifies inherent dynamic stability.
- **$C_{m\delta}$ (Control Moment Effectiveness):** Sensitivity of pitching moment with respect to fin deflection (δ), dictating dynamic maneuverability.

Mathematically, these non-dimensional aerodynamic coefficients are calculated via partial differentiation:

$$C_{m\alpha} = \frac{\partial C_m}{\partial \alpha} \quad (21)$$